

YEAR 9 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION 10.7 - SERIES AND PARALLEL CIRCUITS

Set up the following circuits and answer the questions associated with each one.

INVESTIGATION 10.7

In a line or side by side

AIM To compare series and parallel circuits

Materials:

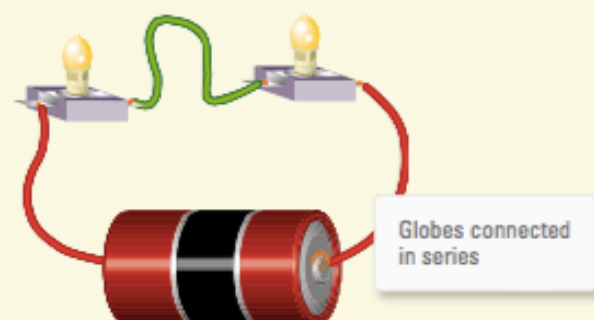
three 2.5-volt or 3.0-volt torch globes

6 connecting leads

Part A: Series circuits

METHOD AND RESULTS

- ▶ Connect one globe and the battery together with wire leads so that the globe lights up.
- ▶ Add a second globe in series with the first globe as shown in the diagram below.
- ▶ Remove one globe from its holder.
- ▶ Replace the globe that was removed, and then remove the other one.



- 1 Draw a circuit diagram to represent the circuit that you have connected.
- 2 How does the brightness of the two globes compare with the brightness of a single globe connected to the same battery?
- 3 What effect does the removal of one globe have on the other globe when the battery is connected?
- 4 Does it matter which globe is removed?

DISCUSS AND EXPLAIN

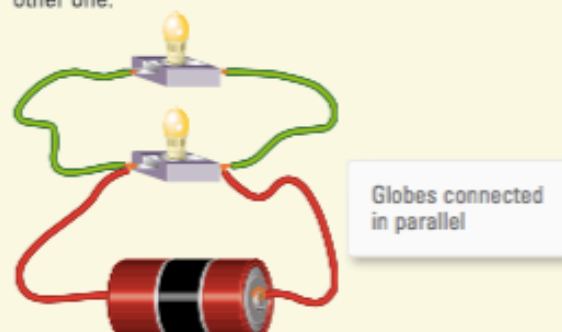
- 5 What would be the effect on the other globes if a third globe were added in series? Test your prediction.
- 6 Can electric current flow in this series circuit when either globe is removed?

- 7 Would it be sensible to have all of the ceiling lights in your home connected in series? Give a reason for your answer.

Part B: Parallel circuits

METHOD AND RESULTS

- ▶ Connect the two globes, battery and wire leads as shown in the diagram below.
- ▶ Remove one globe from its holder.
- ▶ Replace the globe that was removed, and then remove the other one.



- 8 Draw a circuit diagram to represent the circuit that you have connected.
- 9 How does the brightness of the two globes compare with the brightness of a single globe connected to the same battery?
- 10 What effect does the removal of one globe have on the other globe?
- 11 Does it matter which globe is removed?
- 12 Outline whether the removal of one globe has any effect on the other globe.
- 13 What would be the effect on the other globes if a third globe were added in parallel? Design a circuit to test your prediction.

DISCUSS AND EXPLAIN

- 14 Can electric current flow in this parallel circuit when either globe is removed?
- 15 Would it be sensible to have all of the ceiling lights in your home connected in parallel? Give a reason for your answer.